

Message	Signal	Type	Byte	Bits	Factor	Offset	Min
MPPT1_PowerMeas	MPPT1_Vin	S16	0-1	16	0.01	0	-327
MPPT1_PowerMeas	MPPT1_Iin	S16	2-3	16	0.0005	0	-16
MPPT1_PowerMeas	MPPT1_Vout	S16	4-5	16	0.01	0	-327
MPPT1_PowerMeas	MPPT1_Iout	S16	6-7	16	0.0005	0	-16
MPPT1_Status	MPPT1_Mode	U8	0	8	1	0	0
MPPT1_Status	MPPT1_Fault	U8	1	8	1	0	0
MPPT1_Status	MPPT1_Enabled	U8	2	8	1	0	0
MPPT1_Status	MPPT1_AmbientTemp	S8	3	8	1	0	-128
MPPT1_Status	MPPT1_HeatsinkTemp	S8	4	8	1	0	-128
MPPT1_Sweep	MPPT1_SweepIndex	U8	0	8	1	0	0
MPPT1_Sweep	MPPT1_SweepRsvd	U8	1	8	1	0	0
MPPT1_Sweep	MPPT1_SweepCurrent	S16	2-3	16	0.0005	0	-16
MPPT1_Sweep	MPPT1_SweepVoltage	S16	4-5	16	0.01	0	-327
MPPT1_SetMode	MPPT1_ModeCmd	U8	0	8	1	0	0
MPPT1_SweepCmd	MPPT1_SweepStart	S16	0-1	16	0.01	0	-327
MPPT1_SweepCmd	MPPT1_SweepEnd	S16	2-3	16	0.01	0	-327
MPPT1_SweepCmd	MPPT1_SweepSize	U8	4	8	1	0	4
MPPT1_SetVLim	MPPT1_VoltLimitSP	S16	0-1	16	0.01	0	-327
MPPT1_SetILim	MPPT1_CurrLimitSP	S16	0-1	16	0.0005	0	-16
MPPT2_PowerMeas	MPPT2_Vin	S16	0-1	16	0.01	0	-327
MPPT2_PowerMeas	MPPT2_Iin	S16	2-3	16	0.0005	0	-16
MPPT2_PowerMeas	MPPT2_Vout	S16	4-5	16	0.01	0	-327
MPPT2_PowerMeas	MPPT2_Iout	S16	6-7	16	0.0005	0	-16
MPPT2_Status	MPPT2_Mode	U8	0	8	1	0	0
MPPT2_Status	MPPT2_Fault	U8	1	8	1	0	0
MPPT2_Status	MPPT2_Enabled	U8	2	8	1	0	0
MPPT2_Status	MPPT2_AmbientTemp	S8	3	8	1	0	-128
MPPT2_Status	MPPT2_HeatsinkTemp	S8	4	8	1	0	-128
MPPT2_Sweep	MPPT2_SweepIndex	U8	0	8	1	0	0
MPPT2_Sweep	MPPT2_SweepRsvd	U8	1	8	1	0	0
MPPT2_Sweep	MPPT2_SweepCurrent	S16	2-3	16	0.0005	0	-16

MPPT2_Sweep	MPPT2_SweepVoltage	S16	4-5	16	0.01	0	-327
MPPT2_SetMode	MPPT2_ModeCmd	U8	0	8	1	0	0
MPPT2_SweepCmd	MPPT2_SweepStart	S16	0-1	16	0.01	0	-327
MPPT2_SweepCmd	MPPT2_SweepEnd	S16	2-3	16	0.01	0	-327
MPPT2_SweepCmd	MPPT2_SweepSize	U8	4	8	1	0	4
MPPT2_SetVLim	MPPT2_VoltLimitSP	S16	0-1	16	0.01	0	-327
MPPT2_SetILim	MPPT2_CurrLimitSP	S16	0-1	16	0.0005	0	-16
MPPT3_PowerMeas	MPPT3_Vin	S16	0-1	16	0.01	0	-327
MPPT3_PowerMeas	MPPT3_lin	S16	2-3	16	0.0005	0	-16
MPPT3_PowerMeas	MPPT3_Vout	S16	4-5	16	0.01	0	-327
MPPT3_PowerMeas	MPPT3_Iout	S16	6-7	16	0.0005	0	-16
MPPT3_Status	MPPT3_Mode	U8	0	8	1	0	0
MPPT3_Status	MPPT3_Fault	U8	1	8	1	0	0
MPPT3_Status	MPPT3_Enabled	U8	2	8	1	0	0
MPPT3_Status	MPPT3_AmbientTemp	S8	3	8	1	0	-128
MPPT3_Status	MPPT3_HeatsinkTemp	S8	4	8	1	0	-128
MPPT3_Sweep	MPPT3_SweepIndex	U8	0	8	1	0	0
MPPT3_Sweep	MPPT3_SweepRsvd	U8	1	8	1	0	0
MPPT3_Sweep	MPPT3_SweepCurrent	S16	2-3	16	0.0005	0	-16
MPPT3_Sweep	MPPT3_SweepVoltage	S16	4-5	16	0.01	0	-327
MPPT3_SetMode	MPPT3_ModeCmd	U8	0	8	1	0	0
MPPT3_SweepCmd	MPPT3_SweepStart	S16	0-1	16	0.01	0	-327
MPPT3_SweepCmd	MPPT3_SweepEnd	S16	2-3	16	0.01	0	-327
MPPT3_SweepCmd	MPPT3_SweepSize	U8	4	8	1	0	4
MPPT3_SetVLim	MPPT3_VoltLimitSP	S16	0-1	16	0.01	0	-327
MPPT3_SetILim	MPPT3_CurrLimitSP	S16	0-1	16	0.0005	0	-16
MPPT4_PowerMeas	MPPT4_Vin	S16	0-1	16	0.01	0	-327
MPPT4_PowerMeas	MPPT4_lin	S16	2-3	16	0.0005	0	-16
MPPT4_PowerMeas	MPPT4_Vout	S16	4-5	16	0.01	0	-327
MPPT4_PowerMeas	MPPT4_Iout	S16	6-7	16	0.0005	0	-16
MPPT4_Status	MPPT4_Mode	U8	0	8	1	0	0
MPPT4_Status	MPPT4_Fault	U8	1	8	1	0	0
MPPT4_Status	MPPT4_Enabled	U8	2	8	1	0	0
MPPT4_Status	MPPT4_AmbientTemp	S8	3	8	1	0	-128

MPPT4_Status	MPPT4_HeatsinkTemp	S8	4	8	1	0	-128
MPPT4_Sweep	MPPT4_SweepIndex	U8	0	8	1	0	0
MPPT4_Sweep	MPPT4_SweepRsvd	U8	1	8	1	0	0
MPPT4_Sweep	MPPT4_SweepCurrent	S16	2-3	16	0.0005	0	-16
MPPT4_Sweep	MPPT4_SweepVoltage	S16	4-5	16	0.01	0	-327
MPPT4_SetMode	MPPT4_ModeCmd	U8	0	8	1	0	0
MPPT4_SweepCmd	MPPT4_SweepStart	S16	0-1	16	0.01	0	-327
MPPT4_SweepCmd	MPPT4_SweepEnd	S16	2-3	16	0.01	0	-327
MPPT4_SweepCmd	MPPT4_SweepSize	U8	4	8	1	0	4
MPPT4_SetVLim	MPPT4_VoltLimitSP	S16	0-1	16	0.01	0	-327
MPPT4_SetILim	MPPT4_CurrLimitSP	S16	0-1	16	0.0005	0	-16
BPS_SafetyState	VehicleState	U3	0[2:0]	3	1	0	0
BPS_SafetyState	SafetyState	U2	0[4:3]	2	1	0	0
BPS_SafetyState	HV_Ready	U1	0.5	1	1	0	0
BPS_SafetyState	PrechargeActive	U1	0.6	1	1	0	0
BPS_SafetyState	HazardActivate	U1	0.7	1	1	0	0
BPS_SafetyState	BPS_FaultCode	U4	1[3:0]	4	1	0	0
BPS_SafetyState	CockpitFaultRed	U1	1.4	1	1	0	0
BPS_SafetyState	CockpitWarningYellow	U1	1.5	1	1	0	0
BPS_SafetyState	CockpitDCDC_On	U1	1.6	1	1	0	0
BPS_SafetyState	CockpitBattOn	U1	1.7	1	1	0	0
BPS_SafetyState	GraceTimer	U8	2	8	1	0	0
BPS_SafetyState	Relays_B0	U8	3	8	1	0	0
BPS_SafetyState	Relays_B1	U8	4	8	1	0	0
BPS_SafetyState	PrechargeComplete	U1	5.0	1	1	0	0
BPS_SafetyState	VCS_MIA_Detected	U1	5.1	1	1	0	0
BPS_SafetyState	BMS_MIA_Detected	U1	5.2	1	1	0	0
BPS_SafetyState	DeratingPct	U5	5[7:3]	5	4	0	0
BPS_SafetyState	Rsvd6	U8	6	8	1	0	0
BPS_SafetyState	StateAlive	U8	7	8	1	0	0
Vega_SafetyState	VegaState	U8	0	8	1	0	0
Vega_SafetyState	ThrottleOut	U8	1	8	1	0	0
Vega_SafetyState	LHTimer	U8	2	8	1	0	0

Vega_SafetyState	StateAlive	U8	3	8	1	0	0
Vega_ThrottleCMD	Throttle	U8	0	8	1	0	0
Vega_ThrottleCMD	CAN_En	U8	1	8	1	0	0
Vega_ThrottleCMD	SwitchByte	U8	2	8	1	0	0
Vega_ThrottleCMD	Brake_AD	U8	3	8	1	0	0
BPS_PowerLimit	MaxThrottle	U8	0	8	1	0	0
BPS_PowerLimit	MaxRegen	U8	1	8	1	0	0
BPS_PowerLimit	PackV	U16	2-3	16	0.1	0	0
BPS_PowerLimit	PackI	S16	4-5	16	0.1	0	-600
BPS_PowerLimit	SOC	U8	6	8	0.5	0	0
BPS_PowerLimit	LimitReason	U8	7	8	1	0	0
Vega_DriverInputs	PedalRaw	U16	0-1	16	1	0	0
Vega_DriverInputs	BrakeRaw	U16	2-3	16	1	0	0
Vega_DriverInputs	ThrottleOut	U8	4	8	1	0	0
Vega_DriverInputs	RegenOut	U8	5	8	1	0	0
Vega_DriverInputs	Switches	U8	6	8	1	0	0
Vega_DriverInputs	InputAlive	U8	7	8	1	0	0
MC_Status1	Speed	U16	0-1	16	1	0	0
MC_Status1	MotorI	U16	2-3	16	0.1	0	0
MC_Status1	BattV	U16	4-5	16	0.1	0	0
MC_Status1	ErrCode	U16	6-7	16	1	0	0
MC_Status2	ThrottleFB	U8	0	8	1	0	0
MC_Status2	CtrlTemp	U8	1	8	1	-40	-40
MC_Status2	MotorTemp	U8	2	8	1	-30	-30
MC_Status2	DirStatus	U8	4	8	1	0	0
MC_Status2	Switches	U8	5	8	1	0	0
BPS_FaultDetail	FaultCode	U8	0	8	1	0	0
BPS_FaultDetail	TimeSinceOn	U16	1-2	16	1	0	0
BPS_FaultDetail	DetailRsvd	U8	3-6	32	1	0	0
BPS_FaultDetail	DetailAlive	U8	7	8	1	0	0
BPS_SupplBattery	SupplV	U16	0-1	16	0.01	0	0
BPS_SupplBattery	SupplI	S16	2-3	16	0.01	0	-20
BPS_SupplBattery	SupplSOC	U8	4	8	0.5	0	0
BPS_SupplBattery	SupplStatus	U8	5	8	1	0	0
BPS_SupplBattery	SupplTemp	S8	6	8	1	0	-40
BPS_SupplBattery	SupplAlive	U8	7	8	1	0	0
BPS_ChargingInfo	ChargingActive	U1	0.0	1	1	0	0

BPS_ChargingInfo	ChargerConnected	U1	0.1	1	1	0	0
BPS_ChargingInfo	ChgFaultTemp	U1	0.2	1	1	0	0
BPS_ChargingInfo	ChgFaultInput	U1	0.3	1	1	0	0
BPS_ChargingInfo	ChgFaultHW	U1	0.4	1	1	0	0
BPS_ChargingInfo	ChgFaultComm	U1	0.5	1	1	0	0
BPS_ChargingInfo	ChargerOutV	U16	1-2	16	0.1	0	0
BPS_ChargingInfo	ChargerOutI	U16	3-4	16	0.1	-3200	-400
BPS_ChargingInfo	ChargerTemp	S8	5	8	1	-40	-40
BPS_ChargingInfo	ChgInfoRsvd	U8	6	8	1	0	0
BPS_ChargingInfo	ChgInfoAlive	U8	7	8	1	0	0
R_BMS_Fault	DTC1	U16	0-1	16	1	0	0
R_BMS_Fault	DTC2	U16	2-3	16	1	0	0
R_BMS_Fault	Failsafe	U16	4-5	16	1	0	0
R_BMS_Fault	Severity	U8	6	8	1	0	0
R_BMS_Fault	Rolling	U8	7	8	1	0	0
R_BMS_CellExtremes	HiCellIV	U16	0-1	16	0.0001	0	0
R_BMS_CellExtremes	HiCellVID	U8	2	8	1	0	0
R_BMS_CellExtremes	LoCellIV	U16	3-4	16	0.0001	0	0
R_BMS_CellExtremes	LoCellVID	U8	5	8	1	0	0
R_BMS_CellExtremes	CRC16	U16	6-7	16	1	0	0
R_BMS_TempExtremes	HiTemp	S8	0	8	1	0	-40
R_BMS_TempExtremes	HiTempID	U8	1	8	1	0	0
R_BMS_TempExtremes	LoTemp	S8	2	8	1	0	-40
R_BMS_TempExtremes	LoTempID	U8	3	8	1	0	0
R_BMS_TempExtremes	AvgTemp	S8	4	8	1	0	-40
R_BMS_TempExtremes	IntTemp	S8	5	8	1	0	-40
R_BMS_TempExtremes	CRC16	U16	6-7	16	1	0	0
R_BMS_Energy	PackAh	U16	0-1	16	0.1	0	0
R_BMS_Energy	TotalCap	U16	2-3	16	0.1	0	0
R_BMS_Energy	DOD	U8	4	8	0.5	0	0
R_BMS_Energy	PackR	U16	5-6	16	0.001	0	0
R_BMS_Energy	Rsvd	U8	7	8	1	0	0
R_BMS_AdaptEnergy	AdaptSOC	U8	0	8	0.5	0	0
R_BMS_AdaptEnergy	AdaptAh	U16	1-2	16	0.1	0	0

R_BMS_AdaptEnergy	AdaptCap	U16	3-4	16	0.1	0	0
R_BMS_AdaptEnergy	PackOpenV	U16	5-6	16	0.1	0	0
R_BMS_AdaptEnergy	Rsvd	U8	7	8	1	0	0
R_BMS_CellOpenV	LoCellIOV	U16	0-1	16	0.0001	0	0
R_BMS_CellOpenV	LoCellIOVID	U8	2	8	1	0	0
R_BMS_CellOpenV	HiCellIOV	U16	3-4	16	0.0001	0	0
R_BMS_CellOpenV	HiCellIOVID	U8	5	8	1	0	0
R_BMS_CellOpenV	AvgCellIOV	U16	6-7	16	0.0001	0	0
R_BMS_CellResist	LoCellIR	U16	0-1	16	0.01	0	0
R_BMS_CellResist	LoCellIRID	U8	2	8	1	0	0
R_BMS_CellResist	HiCellIR	U16	3-4	16	0.01	0	0
R_BMS_CellResist	HiCellIRID	U8	5	8	1	0	0
R_BMS_CellResist	AvgCellIR	U16	6-7	16	0.01	0	0
BPS_SupplBatt_V	SupplV	U16	0-1	16	0.01	0	0
BPS_SupplBatt_V	SupplI	S16	2-3	16	0.01	0	-20
BPS_SupplBatt_V	SupplSOC	U8	4	8	0.5	0	0
BPS_SupplBatt_V	SupplStatus	U8	5	8	1	0	0
BPS_SupplBatt_V	SupplTemp	S8	6	8	1	0	-40
BPS_SupplBatt_V	SupplAlive	U8	7	8	1	0	0
BPS_DCDC_Stat_V	DCDC_InV	U16	0-1	16	0.1	0	0
BPS_DCDC_Stat_V	DCDC_OutV	U16	2-3	16	0.01	0	0
BPS_DCDC_Stat_V	DCDC_OutI	U16	4-5	16	0.01	0	0
BPS_DCDC_Stat_V	DCDC_Flags	U8	6	8	1	0	0
BPS_DCDC_Stat_V	DCDC_Temp	S8	7	8	1	0	-40
BPS_Emergency	Reason	U8	0	8	1	0	0
BPS_Emergency	SafetyCmd	U8	1	8	1	0	0
BPS_Emergency	Flags	U8	2	8	1	0	0
BPS_Emergency	EmergAlive	U8	3	8	1	0	0
Vega_Fault	FaultCode	U8	0	8	1	0	0
Vega_Fault	SafetyCmd	U8	1	8	1	0	0
Vega_Fault	FaultAlive	U8	2	8	1	0	0
Vega_VehicleStatus	Speed	U16	0-1	16	0.1	0	0
Vega_VehicleStatus	Odo	U24	2-4	24	0.1	0	0
Vega_VehicleStatus	Flags	U8	5	8	1	0	0
Vega_VehicleStatus	LHTimer	U8	6	8	1	0	0
Vega_VehicleStatus	VehAlive	U8	7	8	1	0	0

Max	Unit	Comment
327	V	PV input voltage. Raw: (MSB*256+LSB)*0.01
16	A	PV input current. Raw: (MSB*256+LSB)*0.0005
327	V	Battery output voltage
16	A	Battery output current
6		Operating mode of converter
8		Fault code. Indicated by blinking Limit LED.
1		Converter power stage enabled
127	°C	Ambient temperature sensor
127	°C	Heatsink temp. Derating starts at TempLimitStart.
127	#	IV curve datapoint index (0 to SweepSize-1)
255		Reserved byte
16	A	Current at this IV curve point
327	V	Voltage at this IV curve point
1		Enable/disable MPPT power electronics via CAN
327	V	Sweep start voltage. Limited by max input voltage setting.
327	V	Sweep end voltage
128	#	Number of sweep steps (4-128). MPPT tracks from MPP after sweep.
327	V	Output voltage limit setpoint. For end-of-charge control.
16	A	Output current limit setpoint. For battery charge current control.
327	V	PV input voltage. Raw: (MSB*256+LSB)*0.01
16	A	PV input current. Raw: (MSB*256+LSB)*0.0005
327	V	Battery output voltage
16	A	Battery output current
6		Operating mode of converter
8		Fault code. Indicated by blinking Limit LED.
1		Converter power stage enabled
127	°C	Ambient temperature sensor
127	°C	Heatsink temp. Derating starts at TempLimitStart.
127	#	IV curve datapoint index (0 to SweepSize-1)
255		Reserved byte
16	A	Current at this IV curve point

327	V	Voltage at this IV curve point
1		Enable/disable MPPT power electronics via CAN
327	V	Sweep start voltage. Limited by max input voltage setting.
327	V	Sweep end voltage
128	#	Number of sweep steps (4-128). MPPT tracks from MPP after sweep.
327	V	Output voltage limit setpoint. For end-of-charge control.
16	A	Output current limit setpoint. For battery charge current control.
327	V	PV input voltage. Raw: (MSB*256+LSB)*0.01
16	A	PV input current. Raw: (MSB*256+LSB)*0.0005
327	V	Battery output voltage
16	A	Battery output current
6		Operating mode of converter
8		Fault code. Indicated by blinking Limit LED.
1		Converter power stage enabled
127	°C	Ambient temperature sensor
127	°C	Heatsink temp. Derating starts at TempLimitStart.
127	#	IV curve datapoint index (0 to SweepSize-1)
255		Reserved byte
16	A	Current at this IV curve point
327	V	Voltage at this IV curve point
1		Enable/disable MPPT power electronics via CAN
327	V	Sweep start voltage. Limited by max input voltage setting.
327	V	Sweep end voltage
128	#	Number of sweep steps (4-128). MPPT tracks from MPP after sweep.
327	V	Output voltage limit setpoint. For end-of-charge control.
16	A	Output current limit setpoint. For battery charge current control.
327	V	PV input voltage. Raw: (MSB*256+LSB)*0.01
16	A	PV input current. Raw: (MSB*256+LSB)*0.0005
327	V	Battery output voltage
16	A	Battery output current
6		Operating mode of converter
8		Fault code. Indicated by blinking Limit LED.
1		Converter power stage enabled
127	°C	Ambient temperature sensor

127	°C	Heatsink temp. Derating starts at TempLimitStart.
127	#	IV curve datapoint index (0 to SweepSize-1)
255		Reserved byte
16	A	Current at this IV curve point
327	V	Voltage at this IV curve point
1		Enable/disable MPPT power electronics via CAN
327	V	Sweep start voltage. Limited by max input voltage setting.
327	V	Sweep end voltage
128	#	Number of sweep steps (4-128). MPPT tracks from MPP after sweep.
327	V	Output voltage limit setpoint. For end-of-charge control.
16	A	Output current limit setpoint. For battery charge current control.
4		Vehicle operating state
2		Safety state (TM). LimpHome=90s grace then Failsafe.
1		HV system ready (precharge complete, contactors closed)
1		Precharge sequence in progress
1		Any BPS fault ≠0 → Altair activates hazard flash
13		BPS fault code. Nonzero triggers HazardActivate+CockpitRed.
1		Cockpit RED LED: any BPS fault active
1		Cockpit YELLOW LED: soft limits/derating active, no hard fault
1		Cockpit GREEN LED: DCDC running
1		Cockpit GREEN LED: HV contactors closed
255	s	LimpHome grace timer. Starts 90, counts to 0. At 0 → Failsafe+HV kill.
255		Relay byte 0: contactors + precharge + DCDC
255		Relay byte 1: MPPTs
1		Full precharge sequence complete
1		BPS has detected Vega rolling counter timeout (>500ms)
1		BPS has detected BMS rolling counter timeout
100	%	Current throttle derating percentage (0=full power, 100=zero throttle). 4% steps.
255		Reserved for future use
255		Rolling counter. Vega checks >500ms timeout → BPS_MIA.
255		Packed state byte
255		Actual Kelly_Throttle Vega is sending. BPS can cross-check against MaxThrottle.
255	s	LimpHome countdown from Vega perspective. Should match BPS GraceTimer.

255		Rolling counter. BPS checks >500ms timeout → VCS_MIA → LimpHome.
255		Throttle signal 0-255 (maps 0V-5V)
255		CAN control enable
255		[1:0]=Direction [2]=BrakeSwitch [7:3]=Reserved
255		Brake/regen analog 0-255. Clamped to BPS_PowerLimit.MaxRegen. 0 if regen disabled.
255		Max throttle Vega may send to Kelly. Computed from DCL+SOC+TempDerate.
255		Max regen brake Vega may send to Kelly_Brake_AD. 0=regen disabled. Computed from CCL+HiCellV+SOC.
200	V	Pack voltage from BMS_PackStatus
600	A	Pack current (+discharge -charge). Negative = charging/regen.
100	%	SOC from BMS
255		Limit flags bitmap
4095		Raw pedal ADC (12-bit)
4095		Raw brake analog ADC (12-bit)
255		Computed throttle sent to Kelly_Throttle
255		Computed regen sent to Kelly_Brake_AD. 0 if regen disabled by BPS.
255		Driver switch states packed
255		Rolling counter
6000	rpm	Mechanical speed. (MSB*256+LSB)
400	A	Motor current. ((MSB*256+LSB)/10)
800	V	Battery voltage at MC. ((MSB*256+LSB)/10)
65535		Error bitfield ERR0-ERR15. See KLS CAN Protocol 1.2.
255		Throttle feedback 0-255 (0=0V, 255=5V)
215	°C	Controller temp. Actual = raw - 40
225	°C	Motor temp. Actual = raw - 30
255		Direction status packed
255		Switch/hall sensor states
13		Full BPS fault code for logging/display. All 13 faults.
65535	s	Time since BPS powered on (max ~18hrs)
0		Reserved (future: fault history, timestamps)
255		Rolling counter
40	V	24V lead-acid battery voltage
20	A	Current (+charge -discharge)
100	%	State of charge
255		Battery status
120	°C	Battery temperature
255		Rolling counter
1		Currently charging

1		Battery detected by charger
1		Charger over-temp
1		Input voltage fault
1		Hardware fault
1		CAN comm fault
500	V	Charger output voltage
400	A	Charger output current
150	°C	Charger internal temp
255		Reserved
255		Rolling counter
65535		Relayed DTC1
65535		Relayed DTC2
65535		Relayed failsafe
3		Relayed severity
255		Relayed rolling
5	V	Relayed highest cell V
27	#	Cell ID
5	V	Relayed lowest cell V
27	#	Cell ID
65535		CRC-16 passed through
120	°C	Relayed highest thermistor
18	#	
120	°C	Relayed lowest
18	#	
120	°C	Relayed average
120	°C	Relayed BMS board
65535		CRC-16 passed through
6000	Ah	Relayed amphotours
6000	Ah	Relayed capacity
100	%	Relayed DOD
60	mΩ	Relayed resistance
255		Reserved
100	%	Relayed adaptive SOC
6000	Ah	Relayed adaptive Ah

6000	Ah	Relayed adaptive cap
200	V	Relayed pack open V
255		Reserved
5	V	Relayed lowest OCV
27	#	
5	V	Relayed highest OCV
27	#	
5	V	Relayed average OCV
600	mΩ	Relayed lowest R
27	#	
600	mΩ	Relayed highest R
27	#	
600	mΩ	Relayed average R
40	V	Relayed 24V V
20	A	Relayed current
100	%	
255		
120	°C	
255		
200	V	Relayed HV input
40	V	Relayed 24V output
10	A	Relayed output I
255		[0]=En [3:1]=Fault
120	°C	Relayed temp
255		Shutdown reason code
2		Commanded state
255		Fault flags bitfield
255		Rolling counter
255		VCU fault code
3		[1:0]=SafetyState [3:2]=Severity
255		Rolling counter
150	km/h	Vehicle speed from MC_Speed + wheel diameter
99999	km	Trip odometer
255		System flags
255	s	Limp-Home countdown for telemetry display
255		Rolling counter

0=Disable 1=Enable
0=ConstInputVoltage(Nominal) 1=ConstInputCurrent 2=MinInputCurrent 3=ConstOutputVoltage 4=ConstOutputCurrent 5=TemperatureDerating 6=Fault
0=OK(NoError) 1=ConfigurationError 2=InputOverVoltage 3=OutputOverVoltage 4=OutputOverCurrent 5=InputOverCurrent 6=InputUnderCurrent 7=PhaseOverCurrent 8=Fault
0=Disabled 1=Enabled
0=Disable 1=Enable
0=ConstInputVoltage(Nominal) 1=ConstInputCurrent 2=MinInputCurrent 3=ConstOutputVoltage 4=ConstOutputCurrent 5=TemperatureDerating 6=Fault
0=OK(NoError) 1=ConfigurationError 2=InputOverVoltage 3=OutputOverVoltage 4=OutputOverCurrent 5=InputOverCurrent 6=InputUnderCurrent 7=PhaseOverCurrent 8=Fault
0=Disabled 1=Enabled

0=Disable 1=Enable
0=OFF 1=STANDBY 2=ACCESSORY 4=PRIMARY
0=Normal 1=LimpHome 2=Failsafe
0=NoFault 1=SupplUndervolt 2=SupplOvervolt 3=SupplUndertemp 4=SupplOvertemp 5=DCDC_Undervolt 6=DCDC_Overvolt 7=BPS_Overcurrent 8=CurrentSensorFail 9=PrechargeFault 10=MainPackUV 11=MainPackOV 12=CAN_Fault 13=CAN_Timeout
[0]=HV+ [1]=HV- [2]=MC [3]=MC_Precharge [4]=WC_Relay [5]=DCDC_Precharge [6]=DCDC_Relay [7]=DCDC_On(monitored)
[0]=MPPTA [1]=MPPTA_Pre [2]=MPPTB [3]=MPPTB_Pre [4]=MPPTC [5]=MPPTC_Pre [6]=MPPTD [7]=MPPTD_Pre
[1:0]=SafetyState(0=Normal 1=LH 2=Failsafe) [4:2]=ThrottleState(0=Off 1=Standby 2=Active 3=LimpHome 4=Regen 5=Error) [5]=MC_OK [6]=BPS_OK [7]=HazardReq

1=CAN_Enabled others=Disabled
Direction: 0=Neutral 1=FWD 2=REV 3=Rsvd
b0=OverI b1=OverV b2=UnderV b3=OverTemp b4=LowSOC b5=RegenDisabled b6=LimpHomeActive b7=Reserved
[0]=BrakeSW [1]=ThrottleSW(deadman) [3:2]=DriveMode(0=Park 1=FWD 2=REV) [4]=RegenActive
[1:0]=FeedbackDir(0=Stop 1=FWD 2=REV) [3:2]=CommandDir(0=Neutral 1=FWD 2=REV)
[0]=HallA [1]=HallB [2]=HallC [3]=BrakeSW_12V [4]=RevSW [5]=FwdSW [6]=FootSW [7]=BoostSW
0=NoFault 1=SupplUndervolt 2=SupplOvervolt 3=SupplUndertemp 4=SupplOvertemp 5=DCDC_Undervolt 6=DCDC_Overvolt 7=BPS_Overcurrent 8=CurrentSensorFail 9=PrechargeFault 10=MainPackUV 11=MainPackOV 12=CAN_Fault 13=CAN_Timeout
0=OK 1=Undervoltage 2=Overvoltage 3=Overtemp 4=Disconnected

[illegible]

0=OK 1=UV 2=OV 3=OT 4=Disc
0=OK 1=OT 2=OI 3=UV 4=OV
0=NoFault 1=SupplUndervolt 2=SupplOvervolt 3=SupplUndertemp 4=SupplOvertemp 5=DCDC_Undervolt 6=DCDC_Overvolt 7=BPS_Overcurrent 8=CurrentSensorFail 9=PrechargeFault 10=MainPackUV 11=MainPackOV 12=CAN_Fault 13=CAN_Timeout + 14=BMS_Shutdown 15=VCU_MIA
0=Normal 1=LimpHome 2=Failsafe
b0=CellOV b1=CellUV b2=OverTemp b3=OverCurrent b4=CommFault b5=ContactorFault
0=None 1=PedalSensorFault 2=BrakeSensorFault 3=CAN_Timeout 4=SafetyMismatch 5=MC_MIA 6=BPS_MIA 7=ThrottlePlausibility
State: 0=Normal 1=LH 2=Failsafe Severity: 0=Info 1=Warn 2=Crit 3=Emerg
[1:0]=DriveMode(0=Park 1=FWD 2=REV) [2]=RegenActive [3]=SD_Logging [4]=Tele_Connected